

DPS – FARIDABAD

COMPETITIVE CLASS IX

(i) **Molecular mass (the relative molecular mass) :-** The relative molecular mass (weight) of an element or compound is the mass of one molecule of the element or compound compared with the mass of atom of ${}^6_6\text{C}^{12}$ which is arbitrarily assigned as 12.0000.

(ii) **Gram molecular mass (GMM) :-** The molecular mass expressed in grams is called gram molecular mass (GMM), or gram mole or mole.

1 gram atom of hydrogen = 1.008g

1 gram atom of carbon = 12g

1 gram atom of chlorine = 35.5g

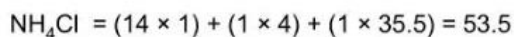
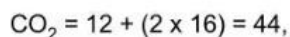
MOLECULAR MASS

It is number of times a molecule is heavier than $1/12$ th of an atom of C^{12} .

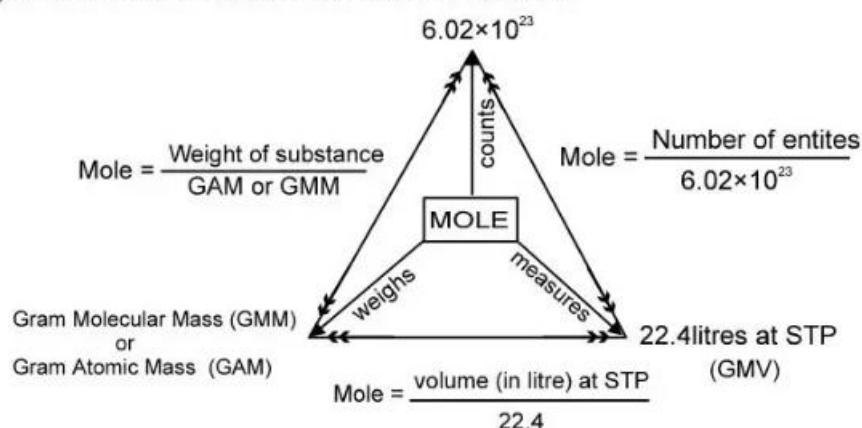
$$\text{Molecular Mass} = \frac{\text{Mass of 1 molecule}}{\frac{1}{12} \times \text{Mass of one atom of } \text{C}^{12}}$$

IMPORTANT POINTS

- ☞ Molecular mass is not a mass (weight) but a number.
- ☞ Molecular mass is relative and not absolute.
- ☞ Molecular mass expressed in grams and is called Gram Molecular Mass (GMM).
- ☞ Molecular mass is calculated by adding all the atomic mass of all the atoms in a molecule.



MASS, MOLE AND NUMBER CONVERSION



(i) Atoms of the same element which have different relative masses are called isotopes.

(ii) In case of isotopes, atomic mass of the elements is average of relative masses of different isotopes of the element.

There are two isotopes of chlorine.

	${}^{35}_{17}\text{Cl}$	and	${}^{37}_{17}\text{Cl}$
relative mass	35		37
relative abundance	3	:	1

$$\text{At. mass of element} = \frac{(\text{At. mass of I isotope} \times \text{relative abundance}_I) + (\text{At. mass of II isotope} \times \text{relative abundance}_{II})}{\text{Total relative abundance}}$$

$$\text{At. mass of Cl} = \frac{(35 \times 3) + (37 \times 1)}{3 + 1} = \frac{105 + 37}{4} = 35.5$$

Ques. Calculate the number of atoms in each of the following

(i) 52 mole of He

(ii) 52 amu of He

(iii) 52g of He

Sol. (i) 1 mole He contain 6.02×10^{23} atoms

52 mole of He contain = $52 \times 6.02 \times 10^{23}$ atoms

(ii) Atomic mass of He = 4 amu

52 amu of He contain = $\frac{52}{4} = 13$ atoms of He

(iii) Number of moles of He in 52g of = 13 moles

number of atoms in 52g of He i.e. 13 moles = $13 \times 6.02 \times 10^{23}$ atoms
= 78.26×10^{23} atoms

SOLVED PROBLEMS

- Ques** Calculate the number of molecules of dinitrogen oxide in 0.044 kg of the gas
(1) 6.02×10^{23} (2) 3.01×10^{23} (3) 12×10^{23} (4) None of these

Ans.

Gram molecular mass of dinitrogen oxide (N_2O) = 44g

Given mass - 0.044kg or 44g

A gram molecular mass of any gas contain Avogadro number of molecules = 6.023×10^{23}

0.044kg of dinitrogen oxide contain 6.023×10^{23} molecules

- Ques** Which of the following contains the least number of molecules and which the highest -
(1) 17.75g of Cl_2 (2) 8g of CO (3) 4g of He (4) 28g of Fe

Ans.

(1) Number of moles of $\text{Cl}_2 = \frac{\text{mass}}{\text{GMM}} = \frac{17.75}{71} = 0.25$

(2) Number of moles of CO = $\frac{\text{mass}}{\text{GMM}} = \frac{8}{28} = 0.285$

(3) Number of moles of He = $\frac{\text{mass}}{\text{GMM}} = \frac{4}{4} = 1$

(4) Number of moles of Fe = $\frac{\text{mass}}{\text{GMM}} = \frac{28}{56} = 0.5$

- Ques** Atomic mass of neon is 20. (i) Calculate the number of atoms in 1g of neon and (ii) 1g atom of neon.
Sol.

(i) 20g of neon contains 6.023×10^{23} atoms

$\therefore$ 1g of neon contains = $\frac{6.02 \times 10^{23}}{20} = 3.01 \times 10^{22}$ atoms

(ii) 1g atom means 1 mole of neon atom, therefore number of neon atoms is 6.02×10^{23}

- Ques** 6×10^{20} molecules of SO_2 are removed from 320 milligram of SO_2 . What are the remaining moles of SO_2 .
(1) 4×10^{-3} moles (2) 5×10^{-3} moles (3) 2×10^{-3} moles (4) 6×10^{-3} moles

Ans. Mole in 320mg. of $\text{SO}_2 = \frac{320 \times 10^{-3}}{64} = 5 \times 10^{-3}$ moles ;

Moles of SO_2 removed are = $\frac{6 \times 10^{20}}{6 \times 10^{23}} = 10^{-3}$ moles

Remaining moles of $\text{SO}_2 = [5 \times 10^{-3} - 10^{-3}] = 4 \times 10^{-3}$ moles

Determination of Chemical Formulae :

Determination of empirical formulae :

Step - I : Determination of percentage

Step - II : Determination of mole ratio

Step - III: Making it whole number ratio

Step - IV : Removal of fractions from mole ratio (to obtain empirical formula)

Determination of molecular formula :

Step - V : Molecular formula = (Empirical formula) n

where $n = \frac{\text{Molecular mass (weight)}}{\text{Empirical mass (weight)}}$

STOICHIOMETRY

- (i) *Stoichiometry* is a Greek word (*stoicheio* = element and *metron* = element)
(ii) Stoichiometry is a calculation of the quantities of reactant and product involved in a chemical reaction.
(iii) Stoichiometry can be classified into two groups -
(a) Gravimetric Analysis (b) Volumetric analysis

Stoichiometry and Problem Solving

In problem solving we shall first discuss gravimetric analysis of chemical reaction. In gravimetric analysis we relate the weights of two substances or a weight of a substance with a volume of a gas or volumes of two or more gases.

Ques.

What amount of MgO is formed when 12g of Mg reacts with oxygen completely.

Sol.

Following are the steps to solve the above problem where mass of reactant is given and mass of product is to be calculated.

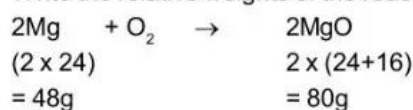
Step 1 Write balance equation



Step 2 Write the moles below the formula



Step 3 Write the relative weights of the reactant and product



Step 4 Apply unitary method

48g of Mg gives 80g of MgO

$$\therefore 12\text{g of Mg gives } \frac{12 \times 80}{48} \text{ g of MgO} = 20\text{g of MgO}$$

By heating 10g CaCO_3 , 5.6g of CaO is formed. What volume of CO_2 obtained in this reaction at STP.

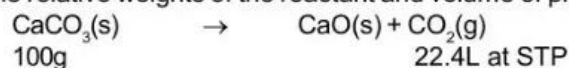
Step 1 Write balance equation



Step 2 Write the moles below the formula



Step 3 Write the relative weights of the reactant and volume of product



Step 4 Apply unitary method

100g of CaCO_3 gives 22.4L of CO_2

$$\therefore 10\text{g } \text{CaCO}_3 \text{ gives } \frac{10 \times 22.4}{100} = 2.24\text{L of } \text{CO}_2 \text{ at STP}$$

Molarity

The number of gram moles of the solute dissolved per litre of the solution. It is denoted by 'M'.

$$M = \frac{\text{Weight}}{\text{Molecular weight}} \times \frac{1}{V (\text{litre})}$$

$$M = \frac{\text{Weight}}{\text{Molecular weight}} \times \frac{1000}{V (\text{ml})}$$

(i) If density and weight percentage of the solution is given then

$$M = \frac{10 \times d \times \text{percent}}{\text{GMM}} \quad \text{where } d = \text{density of solution,}$$

$$\text{Number of millimoles} = M \times V (\text{ml})$$

M/2	M/5	M/10	M/100	M/1000	M	5M	10M
Semi molar solution	Penti molar solution	Deci molar solution	Centi molar solution	Milli molar solution	molar solution	Penta molar solution	Deca molar solution
1/2 mol/L	1/5 mol/L	1/10 mol/L	1/100 mol/L	1/1000 mol/L	1 mol/L	5 mol/L	10 mol/L

Molality :

The number of gram moles of solute dissolved in 1000g or 1kg of the solvent. It is denoted by 'm'.

$$m = \frac{\text{Weight}}{\text{Molecular weight}} \times \frac{1000}{V (\text{ml})}$$

$$m = \frac{\text{Weight}}{\text{Molecular weight}} \times \frac{1}{V (\text{litre})}$$

Mole Fraction :

The mole fraction of a component in a solution is the ratio of the number of moles of that component to the total number of moles present in the solution.

Suppose : – $\left. \begin{array}{l} \text{A – Solute} \\ \text{B – Solvent} \end{array} \right\} \text{Solution}$ n_A = number of moles of solute n_B = number of moles of solvent

$$\text{Then mole fraction of solute} = X_A = \frac{n_A}{n_A + n_B}$$

$$\text{Mole fraction of solvent} = X_B = \frac{n_B}{n_A + n_B} \quad X_A + X_B = 1$$

Dilution Formula $M_1V_1 = M_2V_2$, where M_1 is the initial concentration, V_1 is the initial volume, M_2 is the concentration after mixing or diluting, and V_2 is the total final volume.